Sommersemester 2026

1st Exercise Sheet Solutions

TUTORIUM: Friday, 20.03.2026.

SECTION

01

Getting Familiar with the Density of States

PROBLEM

The calculation of thermodynamic quantities, response functions and Feynman diagrams in QFT for condensed matter systems often requires the evaluation of integrals or sums over all momenta $\boldsymbol{k}$ (typically over the first Brillouin Zone). An important simplification of these $\boldsymbol{k}$-summations is possible, however, when the integrand $\mathcal{F}$ depends on the **energy only**. In this case the integration/sum is best performed by using the energy ε as a variable. In the case of a cubic lattice of volume L^d in d dimensions, for a given observable F, we have:

$$F = \frac{1}{L^d}\sum_{\boldsymbol{k}} \mathcal{F}(\varepsilon_{\boldsymbol{k}}) = \frac{1}{(2\pi)^d}\frac{(2\pi)^d}{L^d}\sum_{\boldsymbol{k}} \mathcal{F}(\varepsilon_{\boldsymbol{k}}) = \int \mathrm{d}\varepsilon\, \mathcal{N}(\varepsilon)\mathcal{F}(\varepsilon),$$

or, for the continuous case,

$$F = \frac{1}{(2\pi)^d}\int \mathrm{d}^d k\, \mathcal{F}(\varepsilon_{\boldsymbol{k}}) = \int \mathrm{d}\varepsilon\, \mathcal{N}(\varepsilon)\mathcal{F}(\varepsilon),$$

where $\mathcal{N}(\varepsilon)$, i.e. the so-called Density of States (DOS), can be defined by comparison of the different expressions as

$$\mathcal{N}(\varepsilon) = \frac{1}{L^d}\sum_{\boldsymbol{k}} \delta(\varepsilon - \varepsilon_{\boldsymbol{k}}),$$

or, for the continuous case,

$$\mathcal{N}(\varepsilon) = \frac{1}{(2\pi)^d}\int \mathrm{d}^d k\, \delta(\varepsilon - \varepsilon_{\boldsymbol{k}}).$$

a)

PROBLEM

Consider the two cases of particles which can move freely and particles whose motion is bound to an infinite lattice with lattice spacing a. Which of the above expressions do you have to use in the first and second case, respectively? Is the integral/summation restricted to certain $\boldsymbol{k}$-vectors? How does the result change, if one considers a one-dimensional, finite lattice (N lattice points, lattice spacing a) with periodic boundary conditions?

SOLUTION

Freely moving particles can have arbitrary momenta, so one integrates over all of momentum space

$$\mathcal{N}(\varepsilon) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \mathrm{d}^d k \, \delta(\varepsilon - \varepsilon_{\boldsymbol{k}}).$$

Particles bound to an infinite lattice with lattice spacing a can only have momenta within the first Brillouin zone

$$\mathcal{N}(\varepsilon) = \frac{1}{(2\pi)^d} \int_{\left[-\frac{\pi}{a}, \frac{\pi}{a}\right]^d} \mathrm{d}^d k \, \delta(\varepsilon - \varepsilon_{\boldsymbol{k}}).$$

Finally, for a one-dimensional finite lattice of length $L = Na$ with periodic boundary conditions, the sum runs over a discrete set of allowed momenta satisfying

$$\mathrm{e}^{\mathrm{i}kL} = 1,$$

i.e. over the N distinct momenta

$$K = \left\{ k \in \left[-\frac{\pi}{a}, \frac{\pi}{a}\right) \mid \mathrm{e}^{\mathrm{i}kNa} = 1 \right\}.$$

Keeping the DOS normalized per unit length, one obtains

$$\mathcal{N}(\varepsilon) = \frac{1}{L} \sum_{k \in K} \delta(\varepsilon - \varepsilon_k) = \frac{1}{Na} \sum_{k \in K} \delta(\varepsilon - \varepsilon_k).$$

If one instead wants the DOS per lattice site, this is simply multiplied by a

$$\mathcal{N}_{\text{site}}(\varepsilon) = \frac{1}{N} \sum_{k \in K} \delta(\varepsilon - \varepsilon_k).$$

b)

PROBLEM

Calculate and plot the explicit expression for $\mathcal{N}(\varepsilon)$ for free, non-interacting particles of mass m (so that $\varepsilon_{\boldsymbol{k}} = \frac{\hbar^2 k^2}{2m}$) in one, two and three dimensions. How do the corresponding Fermi surfaces look like in these cases?

SOLUTION

$$\mathcal{N}(\varepsilon) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \mathrm{d}^d k \, \delta(\varepsilon - \varepsilon_{\boldsymbol{k}}),$$

with $\varepsilon_{\boldsymbol{k}} = \frac{\hbar^2 k^2}{2m}$. Since $\varepsilon_{\boldsymbol{k}} = \frac{\hbar^2 k^2}{2m} \geq 0$, there are no states at negative energy, so the formulas below are understood to vanish for $\varepsilon < 0$. For a given energy $\varepsilon > 0$, the corresponding constant-energy shell is a sphere of radius

$$k_\varepsilon = \frac{\sqrt{2m\varepsilon}}{\hbar}.$$

Using

$$\int_{\mathbb{R}^d} \mathrm{d}^d k\, \delta(g(\boldsymbol{k})) = \int_{g^{-1}(0)} \frac{1}{|\boldsymbol{\nabla} g|}\, \mathrm{d}\sigma(k),$$

which for $g(\boldsymbol{k}) = \varepsilon - \varepsilon_{\boldsymbol{k}}$, $g^{-1}(0) = S^{d-1}(k_\varepsilon)$ and $|\boldsymbol{\nabla} g| = \frac{\hbar^2 k}{m}$ yields

$$\mathcal{N}(\varepsilon) = \frac{1}{(2\pi)^d} \left|S^{d-1}(k_\varepsilon)\right| \frac{1}{\frac{\hbar^2 k_\varepsilon}{m}},$$

where $S^{d-1}(k)$ denotes the radius-k sphere in d dimensions. At $\varepsilon = \varepsilon_{\mathrm{F}}$, k_ε becomes the Fermi momentum k_{F}, and the constant-energy shell becomes the Fermi surface.

- $d = 1$: $|S^0(k)| = 2$, hence

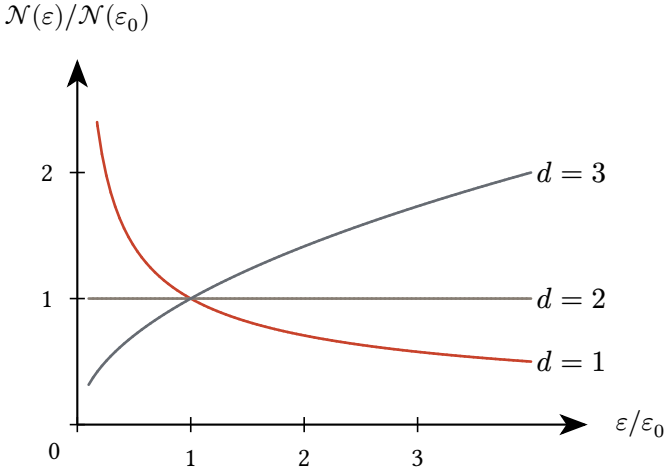
$$\mathcal{N}(\varepsilon) = \frac{1}{2\pi} \frac{\sqrt{2m}}{\hbar\, \sqrt{\varepsilon}}.$$

- $d = 2$: $|S^1(k)| = 2\pi k$, hence

$$\mathcal{N}(\varepsilon) = \frac{1}{2\pi} \frac{m}{\hbar^2}.$$

- $d = 3$: $|S^2(k)| = 4\pi k^2$, hence

$$\mathcal{N}(\varepsilon) = \frac{1}{2\pi^2} \frac{m}{\hbar^2} \frac{\sqrt{2m\varepsilon}}{\hbar}.$$



At $\varepsilon = \varepsilon_{\mathrm{F}}$, these constant-energy shells become the Fermi surfaces: two points in $d = 1$, a circle in $d = 2$, and a sphere in $d = 3$.

c)

PROBLEM

Consider the following one-dimensional tight-binding Hamiltonian

$$H = -t \sum_{\langle i,j \rangle, \sigma} \left(c^\dagger_{i,\sigma} c_{j,\sigma} + \text{h.c.} \right).$$

Here the hopping ($t > 0$) is restricted to neighboring sites, where $c^\dagger_{i,\sigma}$ and $c_{i,\sigma}$ are the creation/annihilation operators for one electron with spin $\sigma = \uparrow, \downarrow$ at the position $x_i = ia$ with $i = 0, 1, ..., N-1$ and a being the lattice spacing. Assuming periodic boundary conditions ($x_0 = x_N$), compute the corresponding eigenenergies, e.g. using the following basis transformation (from real to momentum space)

$$c^\dagger_{k,\sigma} = \frac{1}{\sqrt{N}} \sum_{x_i} \mathrm{e}^{-\mathrm{i}kx_i} c^\dagger_{i,\sigma},$$

for the fermionic operators.

SOLUTION

We start from the given basis transformation

$$c^\dagger_{k,\sigma} = \frac{1}{\sqrt{N}} \sum_{x_i} \mathrm{e}^{-\mathrm{i}kx_i} c^\dagger_{i,\sigma}$$

and invert it to express the real-space operators in terms of momentum-space operators

$$
\begin{aligned}
c_{i,\sigma}^{\dagger} &= \frac{1}{\sqrt{N}} \sum_{k \in K} \mathrm{e}^{\mathrm{i}kx_i} c_{k,\sigma}^{\dagger} \\
c_{i,\sigma} &= \frac{1}{\sqrt{N}} \sum_{k \in K} \mathrm{e}^{-\mathrm{i}kx_i} c_{k,\sigma}.
\end{aligned}
$$

For periodic boundary conditions, the allowed momenta satisfy

$$
\mathrm{e}^{\mathrm{i}kNa} = 1,
$$

i.e.

$$
k = \frac{2\pi}{Na} n, \quad n \in \mathbb{Z},
$$

taken modulo the reciprocal lattice vector $2\frac{\pi}{a}$. Choosing one representative from each equivalence class inside the first Brillouin zone gives the N allowed momenta collected in the set

$$
K = \left\{ k \in \left[-\frac{\pi}{a}, \frac{\pi}{a} \right) \mid \mathrm{e}^{\mathrm{i}kNa} = 1 \right\}.
$$

With this discrete set of momenta, the finite geometric sum in the Hamiltonian becomes the Kronecker delta appearing below. Since $\sum_{\langle i,j \rangle}$ counts each nearest-neighbor bond once, in one dimension we may write it as a sum over j with the neighbor $j+1$ understood modulo N. Substituting this into the Hamiltonian, we obtain (here $j+1$ is understood modulo N because of the periodic boundary conditions)

$$
\begin{aligned}
H &= -t \sum_{j,\sigma} \left(c_{j,\sigma}^{\dagger} c_{j+1,\sigma} + \text{h.c.} \right) \\
&= -t \sum_{j,\sigma} \left(\frac{1}{\sqrt{N}} \sum_{k \in K} \mathrm{e}^{\mathrm{i}kx_j} c_{k,\sigma}^{\dagger} \frac{1}{\sqrt{N}} \sum_{k' \in K} \mathrm{e}^{-\mathrm{i}k'x_{j+1}} c_{k',\sigma} + \text{h.c.} \right) \\
&= -t \frac{1}{N} \sum_{k \in K, k' \in K} \sum_{j,\sigma} \left(e^{\mathrm{i}kja - \mathrm{i}k'(j+1)a} c_{k,\sigma}^{\dagger} c_{k',\sigma} + \text{h.c.} \right),
\end{aligned}
$$

Using the identity

$$
\frac{1}{N} \sum_{j} \mathrm{e}^{\mathrm{i}kja - \mathrm{i}k'(j+1)a} = \mathrm{e}^{-\mathrm{i}k'a} \frac{1}{N} \sum_{j} \mathrm{e}^{\mathrm{i}ja(k-k')} = \mathrm{e}^{-\mathrm{i}k'a} \delta_{k,k'},
$$

we find

$$
\begin{aligned}
H &= -t \sum_{k \in K, k' \in K} \sum_{\sigma} \left(\mathrm{e}^{-\mathrm{i}k'a} \delta_{k,k'} c_{k,\sigma}^{\dagger} c_{k',\sigma} + \text{h.c.} \right) \\
&= -t \sum_{k \in K, \sigma} \left(\mathrm{e}^{-\mathrm{i}ka} c_{k,\sigma}^{\dagger} c_{k,\sigma} + \text{h.c.} \right) \\
&= -t \sum_{k \in K, \sigma} \left(\mathrm{e}^{-\mathrm{i}ka} c_{k,\sigma}^{\dagger} c_{k,\sigma} + \mathrm{e}^{\mathrm{i}ka} c_{k,\sigma}^{\dagger} c_{k,\sigma} \right) \\
&= \sum_{k \in K, \sigma} \varepsilon_k c_{k,\sigma}^{\dagger} c_{k,\sigma}
\end{aligned}
$$

Thus the eigenenergies are

$$\varepsilon_k = -2t\cos(ka)$$

and the corresponding eigenstates are

$$|(k,\sigma)\rangle = c^\dagger_{k,\sigma}|0\rangle.$$

d)

PROBLEM

How can one extend the results of 1c) to arbitrary dimensions $d > 1$? Plot numerically the DOS $\mathcal{N}(\varepsilon)$ for the cases $d = 1, 2, 3$ with $\hbar = m = t = a = 1$. Which are the most prominent features of these DOS functions and at which energies ε do they occur? How would the corresponding Fermi surfaces look like for the cases $d = 1, 2$, e.g. if one has an average density of one electron per site (half-filled system)?

SOLUTION

In arbitrary dimensions $d > 1$, the tight-binding Hamiltonian is

$$H = \sum_{\boldsymbol{k},\sigma} \varepsilon_{\boldsymbol{k}} c^\dagger_{\boldsymbol{k},\sigma} c_{\boldsymbol{k},\sigma},$$

with dispersion

$$\varepsilon_{\boldsymbol{k}} = -2t \sum_{i=1}^{d} \cos(k_i a),$$

obtained by applying the same Fourier transform as in 1c) independently in each lattice direction. The density of states is therefore

$$\mathcal{N}_d(\varepsilon) = \frac{1}{(2\pi)^d} \int_{\left[-\frac{\pi}{a},\frac{\pi}{a}\right]^d} \mathrm{d}^d k\, \delta(\varepsilon - \varepsilon_{\boldsymbol{k}}),$$

For $\hbar = m = t = a = 1$, the band runs from $-2d$ to $2d$.

- $d = 1$: evaluating the delta function at the two roots of $\varepsilon + 2\cos(k) = 0$ gives

  $$\mathcal{N}_1(\varepsilon) = \begin{cases} \frac{1}{\pi\sqrt{4-\varepsilon^2}} & \text{if } |\varepsilon| < 2 \\ 0 & \text{if } |\varepsilon| > 2 \end{cases},$$

  so the DOS diverges at the band edges $\varepsilon = \pm 2$.

- $d = 2$: the square-lattice DOS can be reduced to a complete elliptic integral. In the parameter convention

  $$K(m) = \int_0^{\frac{\pi}{2}} \frac{\mathrm{d}\varphi}{\sqrt{1 - m\sin^2(\varphi)}},$$

  one finds

  $$\mathcal{N}_2(\varepsilon) = \begin{cases} \frac{1}{2\pi^2} K\left(1 - \frac{\varepsilon^2}{16}\right) & \text{if } |\varepsilon| < 4 \\ 0 & \text{if } |\varepsilon| > 4 \end{cases},$$

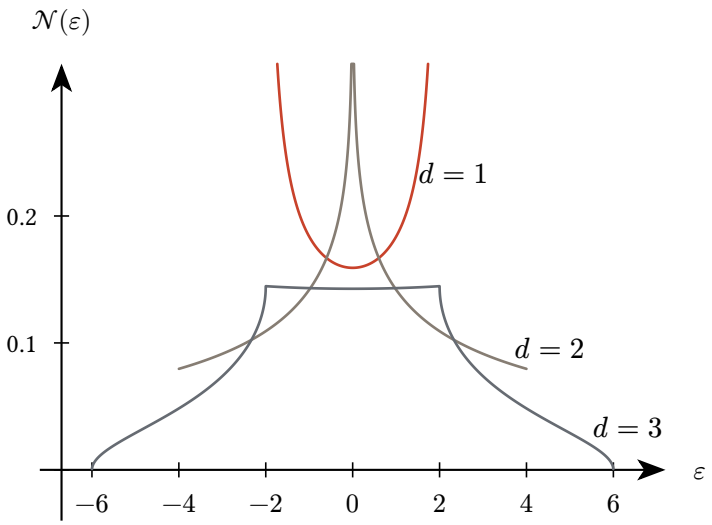
  with a logarithmic van Hove singularity at $\varepsilon = 0$ and the finite band-edge value $\mathcal{N}_2(\pm 4) = \frac{1}{4\pi}$.

- $d = 3$: for the cubic lattice it is convenient to evaluate the DOS numerically via the convolution

$$\mathcal{N}_3(\varepsilon) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \mathrm{d}k_z \, \mathcal{N}_2(\varepsilon + 2\cos(k_z)),$$

  so $\mathcal{N}_3(\varepsilon) = 0$ for $|\varepsilon| > 6$. This DOS remains finite, but develops non-analytic kinks at $\varepsilon = \pm 2$.

The plotted DOS data were generated with a Julia script: it uses the closed-form $d = 1$ result, the elliptic-integral expression for $d = 2$, and the k_z integral above for $d = 3$.



For one electron per site (half filling), exactly half of all available single-particle states are occupied. Since

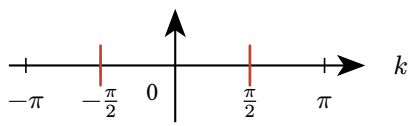
$$\varepsilon_{\boldsymbol{k}} = -2 \sum_{i=1}^{d} \cos(k_i),$$

is particle-hole symmetric on the bipartite hypercubic lattice, the spectrum is symmetric around $\varepsilon = 0$. Therefore at half filling the Fermi energy at $T = 0$ is

$$\varepsilon_{\mathrm{F}} = 0.$$

The Fermi surface is defined by $\varepsilon_{\boldsymbol{k}} = \varepsilon_{\mathrm{F}}$, i.e.

$$-2 \sum_{i=1}^{d} \cos(k_i) = 0.$$



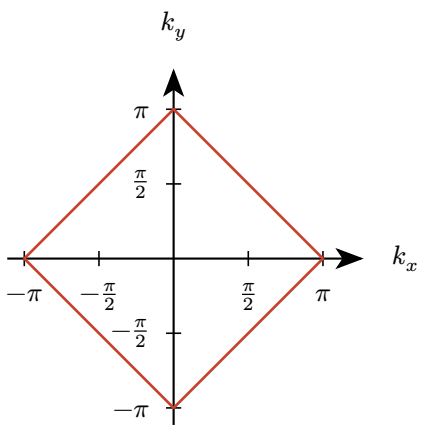
Therefore

$$d = 1: \quad \cos(k) = 0 \quad \rightarrow \quad k = \pm\frac{\pi}{2},$$

and

$$d = 2: \quad \cos(k_x) + \cos(k_y) = 0,$$

i.e. a diamond-shaped Fermi surface in the first Brillouin zone.



SECTION

02

Screened and Unscreened Coulomb Potentials

a)

PROBLEM

From the integral representation of the delta function,

$$\delta(\boldsymbol{r}) = \int \frac{\mathrm{d}^3 k}{(2\pi)^3} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}},$$

and the fact that the Coulomb potential $\phi(\boldsymbol{r}) = -e/r$ satisfies Poisson's equation,

$$-\nabla^2 \phi(\boldsymbol{r}) = -4\pi e \delta(\boldsymbol{r}),$$

show that the electronic pair potential, $V(\boldsymbol{r}) = -e\phi(\boldsymbol{r}) = e^2/r$, can be written in the form

$$V(\boldsymbol{r}) = \int \frac{\mathrm{d}^3 k}{(2\pi)^3} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}} V(\boldsymbol{k}),$$

where the Fourier transform $V(\boldsymbol{k})$ is given by

$$V(\boldsymbol{k}) = \frac{4\pi e^2}{k^2}.$$

SOLUTION

Assume that the Coulomb potential has the Fourier representation

$$\phi(\boldsymbol{r}) = \int \frac{\mathrm{d}^3 k}{(2\pi)^3} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}} \phi(\boldsymbol{k}).$$

Applying $-\nabla^2$ gives

$$-\nabla^2 \phi(\boldsymbol{r}) = \int \frac{\mathrm{d}^3 k}{(2\pi)^3} (-\nabla^2 \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}) \phi(\boldsymbol{k}) = \int \frac{\mathrm{d}^3 k}{(2\pi)^3} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}} k^2 \phi(\boldsymbol{k}),$$

since

$$-\nabla^2 \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}} = k^2 \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}.$$

Poisson's equation also implies

$$-\nabla^2 \phi(\boldsymbol{r}) = -4\pi e \delta(\boldsymbol{r}) = -4\pi e \int \frac{\mathrm{d}^3 k}{(2\pi)^3} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}.$$

Comparing Fourier coefficients gives

$$k^2 \phi(\boldsymbol{k}) = -4\pi e,$$

hence

$$\phi(\boldsymbol{k}) = -\frac{4\pi e}{k^2}.$$

Therefore

$$V(\boldsymbol{k}) = -e\phi(\boldsymbol{k}) = \frac{4\pi e^2}{k^2}.$$

Thus

$$V(\boldsymbol{r}) = \int \frac{\mathrm{d}^3 k}{(2\pi)^3} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}} V(\boldsymbol{k}),$$

with

$$V(\boldsymbol{k}) = \frac{4\pi e^2}{k^2}.$$

b)

PROBLEM

Show that the Fourier transform of the screened Coulomb interaction $V_\mathrm{s}(\boldsymbol{r}) = (e^2/r)\mathrm{e}^{-k_\mathrm{TF} r}$ is

$$V_\mathrm{s}(\boldsymbol{k}) = \frac{4\pi e^2}{k^2 + k_\mathrm{TF}^2},$$

by substituting this expression into the Fourier integral

$$V_\mathrm{s}(\boldsymbol{r}) = \int \frac{\mathrm{d}^3 k}{(2\pi)^3} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}} V_\mathrm{s}(\boldsymbol{k}),$$

and evaluating that integral in spherical coordinates (Hint: The radial integral is best done as a contour integral.). Finally, deduce that $V_\mathrm{s}(\boldsymbol{r})$ satisfies

$$(-\nabla^2 + k_\mathrm{TF}^2) V_\mathrm{s}(\boldsymbol{r}) = 4\pi e^2 \delta(\boldsymbol{r}).$$

SOLUTION

We substitute the Fourier transform of the screened Coulomb interaction into the Fourier integral

$$V_\mathrm{s}(\boldsymbol{r}) = \int \frac{\mathrm{d}^3 k}{(2\pi)^3} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}} V_\mathrm{s}(\boldsymbol{k}) = \int \frac{\mathrm{d}^3 k}{(2\pi)^3} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}} \frac{4\pi e^2}{k^2 + k_\mathrm{TF}^2},$$

and pass to spherical coordinates,

$$V_\mathrm{s}(\boldsymbol{r}) = \int \frac{k^2 \sin\theta \,\mathrm{d}k \,\mathrm{d}\theta \,\mathrm{d}\varphi}{(2\pi)^3} \mathrm{e}^{\mathrm{i}kr\cos\theta} \frac{4\pi e^2}{k^2 + k_\mathrm{TF}^2}.$$

After the trivial φ-integral this becomes

$$V_\mathrm{s}(\boldsymbol{r}) = \int \frac{k^2 \sin\theta \,\mathrm{d}k \,\mathrm{d}\theta}{(2\pi)^2} \mathrm{e}^{\mathrm{i}kr\cos\theta} \frac{4\pi e^2}{k^2 + k_\mathrm{TF}^2}.$$

Since

$$\frac{\mathrm{d}}{\mathrm{d}\theta}\mathrm{e}^{\mathrm{i}kr\cos\theta} = -\mathrm{e}^{\mathrm{i}kr\cos\theta}\mathrm{i}kr\sin\theta,$$

we can write

$$V_\mathrm{s}(\boldsymbol{r}) = \int \frac{k\,\mathrm{d}k\,\mathrm{d}\theta}{(2\pi)^2}\left(\frac{\mathrm{d}}{\mathrm{d}\theta}\mathrm{e}^{\mathrm{i}kr\cos\theta}\right)\frac{1}{-\mathrm{i}r}\frac{4\pi e^2}{k^2+k_\mathrm{TF}^2}$$

and then integrate over θ

$$\int_0^\pi \mathrm{d}\theta\left(\frac{\mathrm{d}}{\mathrm{d}\theta}\mathrm{e}^{\mathrm{i}kr\cos\theta}\right) = \mathrm{e}^{\mathrm{i}kr\cos\pi} - \mathrm{e}^{\mathrm{i}kr\cos 0} = \mathrm{e}^{-\mathrm{i}kr} - \mathrm{e}^{\mathrm{i}kr} = -2\mathrm{i}\sin(kr),$$

so that

$$V_\mathrm{s}(\boldsymbol{r}) = \frac{2e^2}{\pi r}\int_0^\infty \mathrm{d}k\,\frac{k\sin(kr)}{k^2+k_\mathrm{TF}^2}.$$

Since the integrand is even,

$$V_\mathrm{s}(\boldsymbol{r}) = \frac{e^2}{\pi r}\int_{-\infty}^\infty \mathrm{d}k\,\frac{k\sin(kr)}{k^2+k_\mathrm{TF}^2}.$$

To evaluate the radial integral by contour integration, consider

$$I(r) = \frac{e^2}{\pi r}\int_{-\infty}^\infty \mathrm{d}k\,\frac{k\mathrm{e}^{\mathrm{i}kr}}{k^2+k_\mathrm{TF}^2},$$

so that $V_\mathrm{s}(\boldsymbol{r}) = \operatorname{im} I(r)$. The integrand has poles at $k = \pm\mathrm{i}k_\mathrm{TF}$. For $r > 0$, $\mathrm{e}^{\mathrm{i}kr}$ decays in the upper half-plane, so we close the contour there and compute

$$\operatorname*{Res}_{k=\mathrm{i}k_\mathrm{TF}} \frac{k\mathrm{e}^{\mathrm{i}kr}}{k^2+k_\mathrm{TF}^2} = \lim_{k\to\mathrm{i}k_\mathrm{TF}}(k-\mathrm{i}k_\mathrm{TF})\frac{k\mathrm{e}^{\mathrm{i}kr}}{(k-\mathrm{i}k_\mathrm{TF})(k+\mathrm{i}k_\mathrm{TF})} = \frac{\mathrm{i}k_\mathrm{TF}\mathrm{e}^{-k_\mathrm{TF}r}}{2\mathrm{i}k_\mathrm{TF}} = \frac{1}{2}\mathrm{e}^{-k_\mathrm{TF}r}.$$

Hence

$$I(r) = \frac{e^2}{\pi r}2\pi\mathrm{i}\operatorname*{Res}_{k=\mathrm{i}k_\mathrm{TF}}\frac{k\mathrm{e}^{\mathrm{i}kr}}{k^2+k_\mathrm{TF}^2} = \frac{e^2}{r}\mathrm{i}\mathrm{e}^{-k_\mathrm{TF}r},$$

and therefore

$$V_\mathrm{s}(\boldsymbol{r}) = \operatorname{im} I(r) = \frac{e^2}{r}e^{-k_\mathrm{TF}r}.$$

For the second part, Fourier transforming gives

$$(k^2+k_\mathrm{TF}^2)V_\mathrm{s}(\boldsymbol{k}) = 4\pi e^2,$$

which is equivalent to

$$(-\nabla^2+k_\mathrm{TF}^2)V_\mathrm{s}(\boldsymbol{r}) = 4\pi e^2\delta(\boldsymbol{r}),$$

and is indeed satisfied by

$$V_\mathrm{s}(\boldsymbol{k}) = \frac{4\pi e^2}{k^2+k_\mathrm{TF}^2}.$$